

Problem 9.2

If $i^{(6)} = 0.15$, find the equivalent nominal rate of interest convertible semiannually.

Solution

We are given the nominal annual interest rate convertible six times per year:

$$i^{(6)} = 0.15.$$

We want the equivalent nominal annual interest rate convertible semiannually, $i^{(2)}$.

Equivalent nominal rates must produce the same annual effective accumulation. Thus,

$$\left(1 + \frac{i^{(6)}}{6}\right)^6 = \left(1 + \frac{i^{(2)}}{2}\right)^2.$$

Substituting $i^{(6)} = 0.15$ gives

$$\left(1 + \frac{0.15}{6}\right)^6 = \left(1 + \frac{i^{(2)}}{2}\right)^2.$$

Solving for $i^{(2)}$,

$$\begin{aligned}1 + \frac{i^{(2)}}{2} &= \left(1 + \frac{0.15}{6}\right)^{6/2}, \\ \frac{i^{(2)}}{2} &= \left(1 + \frac{0.15}{6}\right)^3 - 1, \\ i^{(2)} &= 2 \left[\left(1 + \frac{0.15}{6}\right)^3 - 1 \right], \\ i^{(2)} &\approx 0.15378125.\end{aligned}$$

Therefore, the equivalent nominal rate convertible semiannually is

$$\boxed{i^{(2)} \approx 0.15378125 = 15.378125\%}.$$